



EVGCPL INTRANET PORTAL

Project Document • Version 6.1 • May 2026

Endpoint Registry • PO_Actual swap • Z12-Driven PCC • Hierarchical Config • Versioned Builds

Portal URL	https://evgcpladmin.github.io/evgcpl-portal/
Repository	https://github.com/EVGCPLAdmin/evgcpl-portal
Current Build	v3.5.0 • build 323 • 2026-05-07 (footer-stamped on every page)
Architecture	12 page HTMLs + portal-bundle.js (821 KB) + portal.css + multi-page-bootstrap.js + /pcc/ subapp
Endpoint Registry	5 logical endpoints — main, pinReset, aiProxy, diagnostic, pcc — overridable from Config
Apps Script files	Router.gs, SafetyHandlers.gs, ScheduledReports.gs, AIChat.gs, AiProxy.gs, SheetDiagnostic.gs, PCHandlers.gs
Auth	PIN + Google OAuth + Vendor/SC modal • Role-based nav restrictions
Scale	64 sites • 333 employees • 319 vendors • 54 SCs • 288 assets
Build script	/build-portal.js — auto-bumps build, patches constants, validates, zips with semver name

1. What's New in v6 (May 2026)

Six major changes shipped since v5, all bundled in this v3.5.0 release:

- Apps Script Endpoint Registry — all /exec URLs now managed in one place via Config →  Apps Script Endpoints. localStorage-backed overrides;  Test buttons; no code changes needed to swap deployments.
- PO Table → PO_Actual — primary tab swap on the Purchase sheet (gid 1458467853). Single PO_TAB constant; 10 fetchSheet call-sites updated.
- Z12-Driven PCC — Nature of Work and Type of Work dropdowns are sourced from Z12.Nature of Work (tab M12_Nature of Work on v2Master). Type filters by parent WBS row's Nature; UoM auto-fills from Z12. Save validates against Z12 vocabulary.
- Workplan rebuilt — per-activity schema (one row per activity, not per month): WBS Code, Nature, Activity, UoM, Qty, Start, End, Duration, % Weight, Responsibility + 4 master-link fields + timestamp.
- Hierarchical Config page — Section accordions with status pills, per-section bulk toggles (All Live / All Off), search filter, role chips. Replaces flat module table.
- Versioned builds — PORTAL_VERSION + PORTAL_BUILD + PORTAL_BUILD_AT constants stamped into the bundle on every package; footer renders v{semver} · build {n} · {date} on every page; zip filename follows the same pattern.

2. Apps Script Endpoint Registry

Five logical Apps Script endpoints are now managed centrally. Defaults are compiled into portal-bundle.js. Runtime overrides are saved to localStorage and applied via the getExec(key) accessor.

Key	Label	Used by
main	Main Backend	Most portal POSTs (DPR, Safety, Onboarding, Reports, generic appendRow/updateCell)
pinReset	PIN Reset	v2_PINReset bound to UserSecrets sheet
aiProxy	AI Proxy (Groq)	EG Assistant chat panel
diagnostic	Sheet Diagnostic	Sharing-Doctor server-side checks
pcc	PCC Handlers	saveProjectSetup, saveBOQ, saveWBS, saveWorkplan, etc.

2.1 How it works

Constants at the top of portal-bundle.js declare the registry shape:

```
const EXEC_REGISTRY_DEFAULTS = { main: { label, desc, defaultUrl }, pinReset: {...},
... };
const EXEC_LS_KEY = 'evgcpl_exec_registry_v1';
function getExec(key) { /* localStorage overrides → defaults */ }
```

All runtime fetches use `getExec(key)` instead of hardcoded literals. Backwards-compatible `APPS_SCRIPT_URL` and `PIN_API_URL` constants still exist but resolve through the same accessor.

2.2 Updating an endpoint without code changes

- Apps Script Editor → Deploy → New deployment (or update existing) → copy the /exec URL
- Portal → Config →  Apps Script Endpoints
- Paste into the right row →  Test → if green, click  Save endpoints
- Reload — the new URL is now live everywhere

2.3 Test mechanics

The Test button POSTs { action: '___ping___' } and surfaces response time + HTTP status as a colored pill. Even "Unknown POST action: ___ping___" is treated as green — it proves the deployment is alive. Only network errors and HTTP 4xx/5xx are red.

3. PO Table → PO_Actual

The primary Purchase Order tab on the Purchase sheet was migrated from 'PO' to 'PO_Actual' (gid 1458467853). The legacy 'PO' tab remains read-only on the same spreadsheet for archive.

3.1 Single source of truth

```
const PO_SHEET_ID = '1zcqF2tjjBETPuW25c9MBMo0zakBIBD6tksg50stFA7c';
const PO_TAB      = 'PO_Actual'; // gid 1458467853 – replaces legacy 'PO' tab as of v3.4.0
```

All fetchSheet calls now reference PO_TAB. To roll back, change one constant — PO_TAB = 'PO' — and the entire portal reverts.

3.2 Updated call sites (10 total)

- SCM dashboard — main grid, KPI strip, recent POs
- Procurement page — PO listing with status filters
- Reports module — PO trend report and PO drill-down
- Notifications scanner — pending PO approval alerts
- Vendor scorecard — PO history per vendor
- Site dashboard — PO totals per site
- Plus 4 secondary readers (analytics, exports, dashboards)

3.3 Sheets registry entry updated

The internal Sheets registry (used by the Config Sheets viewer) was updated:

```
tabs: ['MRS', 'PO_Actual', 'SiteMaster', 'VendorMaster']
```

with notes documenting the gid for future reference.

4. Z12-Driven WBS & Activities (PCC)

After analysing the Planning_v1 AppSheet definition, the PCC architecture was realigned to match how the Planning team already works in AppSheet. Rather than treating M_PL_1_Activities as the strict master, the portal now uses Z12.Nature of Work as the authoritative dropdown source — exactly as AppSheet does.

4.1 Three-tier data model

The model has three roles, each on a different sheet:

Tier	Tab	Role
Master vocabulary	Z12.Nature of Work (M12_Nature of Work on v2Master)	Source of valid (Nature, Type, UOM) triplets — drives every dropdown
Activity catalog	M_PL_1_Activities (on PCC backing sheet)	Cross-project activity catalog — KEY = Nature-Type. Browsable picker.
Per-project rows	WBS / Activities / Workplan	Project-scoped working set written by saveWBS / saveWorkplan

4.2 WBS row

Each WBS row is project-specific. Columns:

- WBS Code — hierarchical (1, 1.2, 1.2.3); editable by user
- WBS Name — description; editable
- Nature of Work — dropdown; values come from DISTINCT(Z12 Nature)

Add WBS Node button creates a blank row with auto-suggested next sequential code (1, 2, 3 ...). User fills the rest.

4.3 Activity row

Each Activity links to a parent WBS. Columns:

- Activity — free-text description; editable
- WBS Code — dropdown of project's WBS rows; picking sets parent and auto-fills Nature
- Nature — read-only, inherited from parent WBS row
- Type of Work — dropdown FILTERED by parent WBS Nature, sourced from Z12 (Nature → Type map)
- Cost Code — editable dropdown from CostCode tab
- UoM — auto-fills from Z12 when Type is picked; editable thereafter
- BOQ Qty — editable number

Two ways to add activities:

- + Add activity — creates a blank row

-  Pick from master — opens M_PL_1_Activities picker for browsing and bulk import; picked rows auto-link to a project WBS row with the same Nature when one exists, otherwise leave WBS Code blank for the user to assign

4.4 UoM auto-fill from Z12

When the user picks a Type of Work on an activity row, the editAct hook looks up Z12 for the (Nature, Type) pair and writes the UOM into the row's Unit field. Implementation:

```
const hint = typeUomHint[nat + '::' + type] || natureUomHint[nat];
```

typeUomHint is keyed by Nature::Type for the most specific match; natureUomHint falls back to the first UOM seen for that Nature in Z12. UoM remains editable after auto-fill so unusual cases can override.

4.5 Save validation

Save() validates against Z12 vocabulary:

- Every WBS row must have a non-empty WBS Code and a Nature in Z12.validNatures
- Every Activity must have a non-empty name and a parent WBS that exists
- If Activity has a Type, it must be in Z12 for that Nature (otherwise stale)
- Activity with no Type triggers a confirm prompt (warning, not block)

4.6 Master Activity Picker

M_PL_1_Activities is still browsable via  Pick from master. The picker:

- Shows all rows by default (was previously project-WBS-gated; that scope is removed)
- Filters: Nature (from Z12), Type (cascades from Nature), WBS Code (DISTINCT from M_PL_1), free text search
- Already-imported rows show as disabled with 'already added' tag
- On confirm, picked rows auto-link to a project WBS row matching the same Nature
- Master link fields (Master UUID, Task Code, CheckSum) carry through for traceability

5. Workplan — Per-Activity Schema

Workplan was rewritten with a one-row-per-activity schema. The previous monthly-grid model (one row per activity per month) was retired in favor of a more flexible Start/End/Duration/Weight model that better fits how activities are actually scheduled.

5.1 Workplan tab — 15 columns

Column	Type	Source / Notes
Project Code	string	Active project context
WBS Code	string	Inherited from activity's WBS
Nature of Work	string	Inherited from WBS's Nature
Activity	string	From master via CheckSum
UoM	string	From master • Unit
Qty	number	BOQ Qty • editable here
Start	YYYY-MM-DD	User-entered
End	YYYY-MM-DD	User-entered
Duration	integer (days)	Auto = (End – Start) + 1
% Weight	number 0–100	Target sum across project = 100
Responsibility	string	Employee Register dropdown
Master UUID	string (FK)	→ M_PL_1_Activities.UUID
Task Code	string	Denormalized from master
CheckSum	string (FK)	→ M_PL_1_Activities.CheckSum
Updated At	ISO timestamp	Server-stamped on every save

5.2 KPIs

- Activities — count linked from WBS
- Total Weight — sum of % Weight, color-coded: green at 100, gold below, red above
- Project Span — earliest Start → latest End across all rows
- Total Duration — sum of activity durations

5.3 Behavior

- Duration auto-computes from Start..End on every cell change
- Responsibility dropdown sourced live from the Employee Register sheet (falls back to free text if unreachable)

- Save shows a confirm prompt if weights don't sum to ~100% (warning, not block)
- '↺ Sync from WBS' re-pulls the activity list from the latest WBS state

6. Hierarchical Config Page

The flat module-and-roles table was replaced with a Section → Module accordion. Each section collapses into a single row showing aggregate state; expanding reveals per-module rows.

6.1 Section accordion

Each of the 11 sections (Main • HR & People • Site Ops • Procurement • Finance • Planning • Plant & Machinery • Reports • Quick Access • Personal • Admin) shows:

- ► chevron + section name (click anywhere to expand)
- State pill: All Live (green) / All Off (grey) / Mixed (amber)
- Module count + per-status chips: '3 live • 1 dev • 1 off'
- Role grants count: '21/35 role grants' for at-a-glance permission density
- Per-section bulk: 'All Live' / 'All Off' buttons that flip the entire section

6.2 Top toolbar

-  Live search — type 'dpr' or 'payment' and only matching modules show; matching sections auto-expand
- Expand all / Collapse all
- Global bulk: All Live / All Off (with confirm)

6.3 Module rows

When expanded, each module is a horizontal grid row with:

- Label + route code (e.g. /budgeting in mono font)
- Status dropdown — color-coded with matching pill background
- Role chips — pill toggles for each role; first-word fits ('MD', 'HR', 'Site'...) with tooltip showing full label
- ↗ Open button — jump to the module immediately

6.4 Self-heal logic

CONFIG_KEY was bumped from evgcpl_portal_config to evgcpl_portal_config_v2. loadPortalConfig now backfills missing routes from MODULE_REGISTRY defaults, so newly-added modules (like Budgeting + Execution) are never silently invisible after a registry update.

7. Versioning & Build Process

7.1 Constants

Three version constants live at the top of portal-bundle.js:

```
const PORTAL_VERSION = '3.4.1';
const PORTAL_BUILD   = 320;
const PORTAL_BUILD_AT = '2026-05-05T18:30:00Z';
```

7.2 Footer

multi-page-bootstrap.js auto-injects #portalVersionFooter on every page load, displaying:

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Fixed-bottom strip with backdrop blur, pointer-events pass through except on the text. Mobile (<600px) drops the date and tail.

7.3 Build script

Run from the repo root:

```
node build-portal.js          // auto-bump build only
node build-portal.js --patch  // 3.4.1 → 3.4.2, build++
node build-portal.js --minor   // 3.4.1 → 3.5.0, build++
node build-portal.js --major   // 3.4.1 → 4.0.0, build++
node build-portal.js --version=4.0.0 // explicit semver, build++
```

The script:

- Reads version.json
- Applies the bump and updates the build timestamp
- Patches the three constants in portal-bundle.js
- Runs node --check to validate JS syntax
- Saves the new version.json
- Creates EVGCPL_Portal_v{semver}_build{n}_{YYYYMMDD}.zip

7.4 Bump conventions

Trigger	Bump
Bug fix or small UX tweak	--patch (3.4.0 → 3.4.1)
New module / route / section ships	--minor (3.4.x → 3.5.0)
Architecture change (e.g. v2 single-file → v3 multipage)	--major (3.x.x → 4.0.0)
Every package, regardless of bump	build++ (always increases monotonically)

8. Apps Script Backend

All server-side logic lives in a Google Apps Script project bound to the master spreadsheet, deployed as a Web App at the exec URL. Files in /apps-script/ in the repo are reference copies — they need to be manually pasted into the Apps Script Editor and a new version deployed for changes to be live.

8.1 File map

File	Purpose
Router.gs	doPost / doGet entry points + action dispatch table. Every route is documented here.
SafetyHandlers.gs	Generic write API: appendRow, updateCell, batchUpdate. Plus closeSafetyIncident.
ScheduledReports.gs	Time-driven email triggers; saveScheduledReport, runReportNow.
AIChat.gs	aiChat action — Groq llama-3.3-70b-versatile via Apps Script proxy.
AIProxy.gs	Generic LLM proxy with Gemini fallback (uses GEMINI_API_KEY if Groq unavailable).
SheetDiagnostic.gs	diagnoseSheet, listShares — server-side sheet sharing checks.
PCCHandlers.gs	10 PCC actions: saveProjectSetup, saveBOQ, saveWBS, saveWorkplan, saveManpower, saveMachinery, saveMaterials, saveOverheads, saveVariations, submitBudgetApproval.
README.md	Install/deploy instructions + full action map + common gotchas.

8.2 Deploy cycle

After editing any .gs file:

- Save in the Apps Script Editor
- Deploy → Manage deployments →  on the active deployment
- Version: New version → Deploy
- The exec URL stays the same. The new code is live within seconds.

Editing a .gs file alone is NOT enough — Apps Script Web Apps cache the deployed snapshot. Redeploying a new version is mandatory.

8.3 Required Script Properties

- GROQ_API_KEY — gsk_... key from console.groq.com (free tier: 30 req/min)
- GEMINI_API_KEY — optional fallback for AI when Groq is down

9. Sheet IDs & Constants

Constant	Sheet ID	Purpose
Z12_TAB NEW	'M12_Nature of Work' on V2_MASTER	Drives Nature/Type/UoM
EXEC_REGISTRY NEW	5 logical Apps Script endpoints	Runtime overrides via Config
SHEET_ID (Master)	1B2wb38KhNwlLoZnsAGWQkO0FdEGFFfsh3ycRRurigq4	BillingMaster, SiteMaster
EMP_SHEET_ID	1HWKZPhKRhcuvxBgyyN8zRt8p-SzYmKjJWiOdCgykBHs	Employee Register
V2_MASTER_SHEET_ID	1fhSO4WBYP0LNXpXe9I9zr5qsIPs9CIDFpUixBogPnsM	v2 Master • hosts Z12 (M12_Nature of Work)
DPR_SHEET_ID	139deMPqCXVZLSw5gFdzhN1heV-hynkYqljsVpO1oXSc	Daily Progress Reports
PO_SHEET_ID	1zcqF2tjjBETPuW25c9MBMo0zakBIBD6tksg5OstFA7c	Purchase / PO
PO_TAB NEW	'PO_Actual' (gid 1458467853)	Replaces legacy 'PO' tab
PAYMENT_SHEET_ID	1mLddxLRf719EaXE9XSET9gT8l0a8Cxns362ylbHo63g	Payments
STORES_SHEET_ID	1iMQxgqGilUh2_3NCZI5D-EMt-NC8FwugX83q2fWb8fE	Stores • StockIN • GRN
SAFETY_SHEET_ID	1B8P0PawV43ksazbzhKsil1X6-INOfxx9PFvGycNOvDY	Safety • Incidents
REWARDS_SHEET_ID	1vz8HLopjISF8TF7rzYuVu5JjqukT929I7a5x7kdehll	Rewards & Wall
PIN (UserSecrets)	1hN4VEDNpVLD3lKuBPYCTOaViv7UpveRfud2d2gy15D0	UserSecrets
PCC backing sheet	1dQow9nD4e0qVOSfpwEWQmPTuhF3FW_8r1oK5dMjJIRE	PCC + M_PL_1_Activities
BUDGET_SHEET_ID	(empty — pending Drive upload)	IC Budget template

10. Open Points & Pending Actions

10.1 User actions required

#	Item	Notes
1	Deploy Apps Script Router.gs	Paste Router.gs into the bound Apps Script project to wire the 30+ actions. Without this, PCC saves return 'Unknown POST action'.
2	Paste PCCHandlers.gs	Sibling file to Router.gs that contains the 10 PCC handler functions
3	Redeploy Web App	Deploy → Manage deployments →  → New version → Deploy. Exec URL stays the same.
4	Confirm PO_Actual sharing	Sheet 1zcgF2tjj.../gid 1458467853 must be shared 'Anyone with the link' for portal reads
5	Verify GROQ_API_KEY	Apps Script → Project Settings → Script Properties. AI Assistant is unresponsive without this.
6	Test all endpoints after deploy	Portal → Config →  Apps Script Endpoints →  Test all (should be 5 green pills)
7	Upload IC Budget Excel	Set BUDGET_SHEET_ID for the Afcons-format IC Budget template (separate from PCC)

10.2 Dev pending

- Grade_MatKey composite column in 10_Mix_Design_Lookup (HIGH)
- AppSheet CSV export with Virtual Columns (MED — awaiting clarification on target app)
- My Team reportees card debug — empCode vs managerCode value mismatch (MED)
- Onboarding checklist — final item list pending HR confirmation (MED)
- Payslip PDF generation — phase 3 payroll integration (LOW)
- SC Work Orders / Job Cards — sheet not yet available (LOW)

10.3 Architecture / infrastructure

- CNAME: intranet.evgcpl.com → not planned for now
- OAuth External audience → not required (PIN + Google + Vendor modal cover all)
- Repo URL update → 'evgcpladmin' → 'EVCPLAdmin' (use git remote set-url)

11. Build Log

Version	Date	Key Changes
v1.0	Jan 2026	Initial portal — Dashboard, HR, Accounts, SCM. Sidebar nav. PIN auth. gviz fetch.
v2.0	Feb 2026	Top nav bar. Rewards & Wall. AppSheet/Sheets split. Safety. Site Manager.
v3.0	Mar 2026	Onboarding. My Profile. My Team. Scheduled emails. Planning skeleton.
v4.0	Apr 2026	My Profile redesign. Per-report scheduling. Apps/Sheets split. Planning iframe.
v5.0	May 2026	PCC subapp integrated. Planning → Budgeting + Execution. Dashboard + Command Center merge for MD. DPR sheet wired.
v6.0	May 2026	Apps Script Endpoint Registry • PO → PO_Actual swap • Workplan rebuilt (per-activity schema) • Hierarchical Config page • Versioned builds with footer stamping.
v6.1	May 2026	Z12-Driven PCC: Nature/Type/UoM dropdowns now sourced from Z12.Nature of Work (M12_Nature of Work on v2Master). WBS picker dropped (project-specific rows). Activity Type filtered by parent WBS Nature. UoM auto-fills from Z12. Master Activity Picker still browsable for M_PL_1_Activities catalog import. Bundle: v3.5.0 build 323.

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